Complex Analysis of Askaryan Radiation: UHECR Reconstruction with Askaryan Radio Array

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Ultra-high energy cosmic rays (UHECR) can produce relativistic cascades that emit radio-frequency (RF) pulses in the 10-1000 MHz bandwidth via two distinct effects: the geomagnetic effect, and the Askaryan effect. The geomagnetic effect occurs when the magnetic field of the Earth causes cascade charges to form a transverse current that radiates linearly polarized radiation aligned with the Lorentz force direction. The Askaryan effect is caused by the net negative charge excess in the cascade that radiates linearly polarized radiation along the Cherenkov cone. When UHECR cascades enter solid, RF transparent matter at altitudes where the cascade develops, Askaryan radiation can propagate through the solid matter to RF detectors. The Askaryan Radio Array (ARA) at the South Pole has observed 13 UHECR candidates in precisely this fashion. We present an analytical model that confirms the events are UHECRs. The model includes the Askaryan effect and the ARA RF channel response. The coherently summed waveforms (CSWs) from the UHECR candidates match our model with correlation coefficients between 0.69 and 0.86, and with minimal fractional power differences. From the fit between model and data, we extract the Askaryan pulses.

Keywords: Ultra-high energy neutrino; Askaryan radiation; Mathematical physics

I. INTRODUCTION

UHECR cascades produce RF pulses via the geomagnetic and Askaryan effects, and these pulses have been used to study UHECR properties [1, 2]. Detectors with RF channels sensitive to RF pulses from UHECRs add complementary data for event reconstruction to traditional ground-based muon and fluorescence detectors. UHECR candidate events have now been presented by the Antarctic Ross Ice Shelf Antenna Neutrino Array (ARIANNA), ARA, and Radio Neutrino Observatory Greenland (RNO-G) collaborations [3–5]. In ARA, the contribution from the Askaryan effect dominates the observed RF pulses. Due to the unique location of ARA at the South Pole, the contribution from the Askaryan effect dominates the observed RF pulses. The South Pole is 2800 meters above sea level, and the thin atmosphere allows UHECR cascades to develop near the altitude of the ARA RF detection channels. As charged particles from the UHECR cascade enter the ice, the negative excess charge radiates collectively in the 0.1-1 GHz bandwidth.

Within the genre of calculations that predict the properties of RF pulses from UHECRs and ultra-high energy neutrinos (UHE-ν), there are three classes of models. First, Monte Carlo programs may be used to track each particle and corresponding RF emission within the cascade. These are known as full Monte Carlo, or *full-MC calculations* [6–8]. Second, there are formalisms in which the analytic vector potential corresponding to the radiating charge is convolved with the simulated cascade profile. These are called *semi-analytic calculations*, and have been used to predict the RF pulses from UHE-ν interacting in Antarctic and Greenlandic ice [9–11]. Finally, there are *fully analytic* calculations that derive the UHECR or UHE-ν signal from classical electrodynamics. Examples of such models are Ralston and Buniy (RB2001) [12], Hanson and Connolly (HCRB2017) [13], and Hanson and Hartig (HHH2022) [14].

One key advantage of fully analytic Askaryan calculations is that they can be fit to raw data from Askaryan-class detectors. A prediction for the voltage trace from an RF detection channel is produced by convolving the analytic Askaryan model with an accurate model of the RF channel response. Analytic voltage traces can be matched to the observed voltage traces by maximizing the correlation coefficient from cross-correlation while tuning parameters like the Askaryan pulse width. Recently, Hanson and Hartig presented evidence that the HH2022 model, when convolved with an RF channel model, produces analytic voltage traces that match voltage traces from UHE-ν signals in NuRadioMC [11]. This technique can be used to reject several years of thermal noise backgrounds in detectors like ARA and RNO-G (HH2026) [15]. Further, once the match between observed and analytic voltage trace is optimized, the Askaryan electromagnetic pulse can be extracted based on the fit parameters. In this work, we show that the HH2026 procedure matches the voltage traces from 13 UHECR candidate events observed by ARA [4].

The remaining sections of this work are organized as follows: in Sec. II, we define units, functions, operations on functions, and notational conventions. In Sec. III, we review the design of the Askaryan Radio Array detector, including the properties of the RF channels. In Sec. IV, we present the match between our theoretical voltage traces and the ARA data. Finally, in Sec. V, we summarize our results and indicate future directions of the research.

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II. UNITS, DEFINITIONS, AND CONVENTIONS

Our goal is to demonstrate a match between a theoretical voltage trace for a UHECR candidate in ARA RF channels and the observed voltage traces from those same channels. Let the Askaryan *signal* created by the portion of the UHECR cascade radiating in the ice be $s(t)$. Let E_0 be the amplitude in V m^{-1}, t be the time in ns relative to peak emission, and σ_t be the pulse width in ns. According to Eqs. 28 and 9 and from [14, 15], respectively, $s(t)$ is

$$s(t) = -E_0 t e^{-\frac{1}{2}(t/\sigma_t)^2} \tag{1}$$

To predict a voltage trace from an RF dipole channels in ARA, $s(t)$ must be convolved with $r(t)$, the RF channel response. Let R_0 be the amplitude of the response, with units of m ns^{-1}, f_0 be the resonance frequency of the dipole in GHz, and γ be the exponential decay constant in GHz. According to Eq. 31 from [15], $r(t)$ is

$$r(t) = R_0 e^{-2\pi\gamma t}\cos(2\pi f_0 t) \tag{2}$$

The predicted voltage trace is given by the convolution of $s(t)$ and $r(t)$, denoted $s(t) * r(t)$. The convolution is defined as:

$$s(t) * r(t) = \int_{-\infty}^{\infty} s(t-\tau) r(\tau) d\tau \tag{3}$$

Let $u(t)$ represent the Heaviside step function. According to Eq. 55 from [15], $s(t) * r(t)$ becomes

$$s * r = -E_0 R_0 \int_{-\infty}^{\infty} (t-\tau) e^{-\frac{1}{2}\left(\frac{t-\tau}{\sigma_t}\right)^2} \Re\left\{e^{2\pi j f_0 \tau} e^{-2\pi\gamma\tau}\right\} u(\tau) d\tau \tag{4}$$

The Heaviside step function is introduced to ensure causality. When the voltage trace is predicted with $s(t) * r(t)$, the result must be matched to observed traces from the RF channels with fixed, measured values for γ and f_0. The match between $s(t) * r(t)$ and $d(t)$ is achieved by tuning σ_t to maximize the peak value of the cross-correlation. For real signals $f(t)$ and $g(t)$, cross-correlation is defined as

$$f(t) \star g(t) = \int_{-\infty}^{\infty} f(\tau) g(t+\tau) d\tau \tag{5}$$

The algorithm used to compute the cross-correlation in this analysis is `scipy.signal.correlate`. To maximize the signal to noise ratio (SNR) of UHECR data, traces from multiple channels are combined to form the *coherently summed waveform* (CSW). The CSW is computed from a set of traces using the following algorithm: (1) select a reference trace, (2) cross-correlate with the next trace in the set until the cross-correlation is maximized, (3) add the two traces to form a new reference, and (4) repeate until each trace in the set is added coherently to the reference. The CSW is normalized so that the inner product of the trace is one. The theoretical trace is normalized in the same way. Normalized traces are unitless but retain nanoseconds for the time unit.

The solution to Eq. 4 was shown in Eq. 61 of [15]. Let $j = \sqrt{-1}$, $x = t/(\sqrt{2}\sigma_t)$, $z = (2\pi j f_0 - 2\pi\gamma)\sqrt{2}\sigma_t$, $w(z)$ be the Faddeeva function of a complex variable, and

$$q = -j\left(x + \frac{z}{2}\right) \tag{6}$$

The solution to Eq. 4 is

$$s * r = -\sqrt{\pi} R_0 E_0 \sigma_t^2 \left(x e^{-x^2} \Re\{w(q)\} - \frac{1}{2} e^{-x^2} \Re\left\{-j\frac{dw(q)}{dq}\right\}\right) \tag{7}$$

A Python3 code was given in [15] to compute Eq. 7. Note that q depends on x, which depends on t. The $s * r$ oscillates at a frequency $\approx f_0$ because the Faddeeva function oscillates with a complex argument. The envelope of $s * r$ is limited by the width of $s(t)$ and the exponential decay of $r(t)$.

III. THE ASKARYAN RADIO ARRAY

The Askaryan Radio Array consists of five stations deployed near the South Pole designed to detect UHE-ν and UHECR events via the Askaryan effect in Antarctic ice [16–19]. The A5 station utilizes a dedicated RF trigger on the CSW output of a phased array of seven vertically polarized dipoles. The RF phased array trigger boosts the SNR of UHE-ν and UHECR events, and allows for increased sensitivity to rare fluxes of these particles [20]. The ARA UHECR candidates used in this analysis were recorded on the A5 station via the RF phased array trigger. A schematic of the A5 detector is shown in Fig. 1, and the A5 detector parameters are summarized in [16].

The UHECR candidate events were detected with the RF phased array trigger in A5. The proposed event geometry is shown in Fig. 2. The UHECR candidates would normally create both geomagnetic and Askaryan radiation, and geomagnetic radiation usually dominates the total electromagnetic pulse when the detector lies far beneath the cascade [3]. The same claim is made in recent UHECR candidate detections via RNO-G [5], though RNO-G and ARA are situated at high altitudes. At the South Pole, however, geomagnetic radiation is expected to be horizontally polarized due to the vertical

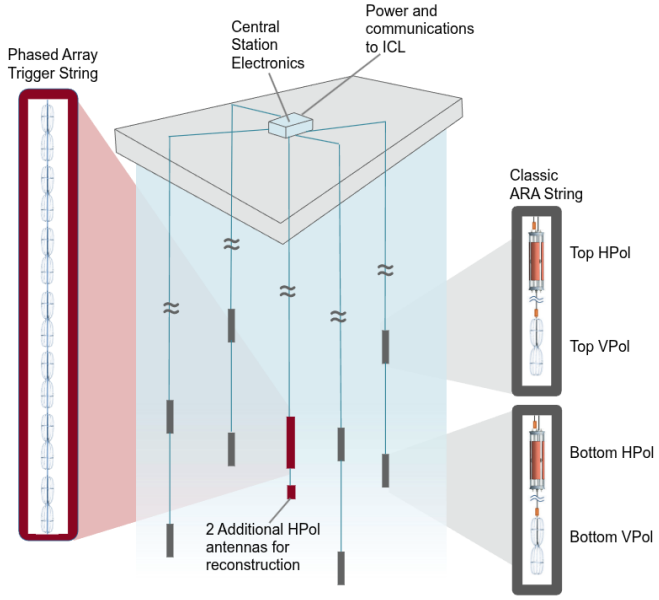

FIG. 1: A general schematic of the A5 station within the Askaryan Radio Array (ARA) detector. The RF phased array trigger string is in the center, surrounded by four reconstruction strings. Strings have both HPol and VPol RF channels. Figure adopted from [16].

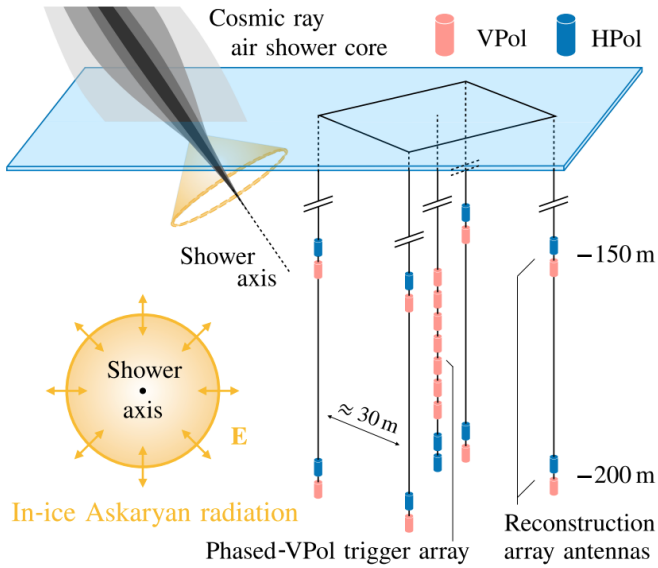

FIG. 2: The proposed UHECR candidate event geometry consists of a high-energy cascade interacting along the shower axis that enters the ice and creates Askaryan radiation. Figure adopted from [4].

orientation of the local magnetic field. The vertically polarized (VPol) RF channels are therefore less sensitive to geomagnetic radiation. Data from the horizontally polarized (HPol) RF channels is not included in our analysis. For the VPol phased array channels, we will show in Sec. IV that geomagnetic radiation is not necessary to explain the observed voltage traces.

IV. RECONSTRUCTION ANALYSIS

Our cross-correlation analysis between Eq. 7 and the CSWs from the UHECR candidates from A5 begins by making two assumptions. The first is that the f_0 and γ parameters are identical across the A5 phased array trigger string. While each channel may not have *identical* parameters, that is the intent of the design shown in Fig. 1. The second assumption we make is that calibration pulse events resemble Askaryan pulses. Though that is the purpose of the calibration pulser system design in ARA [16, 18], the original pulser voltage is transmitted through an ARA dipole, propagates to the A5 trigger string, and is converted to the observed trace. None of these effects, however, should alter f_0 or γ, as they are properties of the RF antennas themselves.

Given these assumptions, we measured f_0 and γ for the A5 trigger string using a data set of events triggered by the A5 calibration pulser. Given that the lower edge of the RF channels is of order ≈ 0.1 GHz, and that calibration pulser events are of order $\lesssim 100$ ns wide, we hypothesized that $f_0 \approx 0.1$ GHz and $\gamma \approx 0.01$ GHz. Given these

ID	σ_t (ns)	ρ
1995-0039	0.6 ± 0.1	0.795
2250-1064	0.6 ± 0.1	0.802
2250-1072	0.6 ± 0.1	0.767
2250-1081	0.6 ± 0.1	0.776
2250-1088	0.6 ± 0.1	0.810
2250-1107	0.6 ± 0.1	0.816
2250-1119	0.6 ± 0.1	0.770
	Average:	0.79 ± 0.02
Best fit f_0, γ [GHz]	0.19 ± 0.01	0.02 ± 0.01

TABLE I: Results for f_0 and γ from calibration pulser data.

initial guesses, we calculated the CSW for each event, and maximized the correlation coefficient for each CSW, ρ, by tuning σ_t. We scanned the $[f_0, \gamma]$ parameter space until the global average ρ_{ave} was maximized. We found that $\rho_{\text{ave}} = 0.79 \pm 0.02$ was maximized when $f_0 = 0.19 \pm 0.01$ GHz, and $\gamma = 0.02 \pm 0.01$ GHz. Though σ_t was allowed to vary for each CSW, the maximum ρ was reached when $\sigma_t = 0.6 \pm 0.1$ for each CSW individually. This is not a coincidence, but encouraging, for the calibration pulser properties tend to remain constant over time. These results are summarized in Tab. I.

Several points about the calibration pulser events are worth mentioning. First, the pulse from the calibration pulser is transmitted through an RF dipole antenna, and received with the A5 trigger string. The received voltage traces are technically the derivative of the original pulse, convolved twice with the effective height of the RF dipole (see Eq. 9 of [21]). For identical RF dipole transmitter

and receiver, however, the parameter f_0 will not change due to the double convolution. Thus, fitting our model of the Askaryan pulse convolved with a single copy of the RF dipole response function will still oscillate with frequency f_0. As a check, we fit the UHECR data while scanning σ_t, f_0, and γ as free-parameters. The result for f_0 was 0.19 ± 0.01 GHz, remarkably consistent with the result in Tab. I. The result for γ was 0.015 ± 0.02 GHz, also consistent with Tab. I. The larger fractional error in γ occurs because the effect of an error in γ on the overall pulse width can be mitigated by a larger or smaller σ_t.

After measuring f_0 and γ, we proceeded to analyze the UHECR candidate voltage traces. We computed the CSW for each UHECR candidate event, and maximized ρ by only tuning σ_t, given our results for f_0 and γ in Tab. I. The CSWs were found in most cases to contain a reflected pulse. In this context, a reflected pulse is a delayed copy of the primary pulse with a smaller amplitude. For each UHECR candidate CSW, we performed a fit assuming two copies of our model. Let the primary pulse width be $\sigma_{t,1}$, the reflection width be $\sigma_{t,2}$, and let Δt be the delay between the pulses. We scanned over the $[\sigma_{\mathrm{t},1}, \sigma_{\mathrm{t},2}, \Delta t]$ parameter space to maximize ρ for the CSW. Initially, we set the amplitude of the reflection to 10% relative to the primary pulse. After obtaining the best-fit results for $[\sigma_{\mathrm{t},1}, \sigma_{\mathrm{t},2}, \Delta t]$, we then tuned the relative amplitude of the reflection to minimize the power difference ΔP between CSW and model. We define the ΔP between waveform data samples d_i and the model waveform samples m_i like

$$\Delta P = \sum_i (d_i - m_i)^2 \tag{8}$$

The definition is loosely based on the power of an AC voltage, like $P = V^2/R$. In our best fits for CSWs, we quote the reflection amplitude that minimizes ΔP.

The best-fit analytic traces and UHECR candidate CSWs are shown in Figs. 3-6, and the numerical results for $\sigma_{\mathrm{t},1}$, $\sigma_{\mathrm{t},2}$, Δt, relative amplitude, and ρ are shown in Tab. II. The same set of results, with relative amplitude set to zero, are shown in Tab. III. Reconstructed primary pulse widths, $\sigma_{\mathrm{t},1}$, tend to be < 1 ns, while reflected pulse widths, $\sigma_{\mathrm{t},2}$, tend to be > 1 ns. The reflected pulses are delayed by ≈ 25 ns, with two exceptions. For event ID 1957-13330, the reflected pulse occurs within the envelope of the primary pulse, altering the total envelope (see Fig. 3 top right). For event ID 2716-58611, there is no reflected pulse that can be fit by our algorithm. The errors on σ_t, Δt, and relative amplitude are limited by our scan resolution. Note that the sampling rate for the traces was 1.5 GHz. Because 1/1.5 GHz ≈ 0.67 ns, we do not trust our timing precision far below 0.67 ns.

The results for $\sigma_{\mathrm{t},1}$ in Tab. III are similar to those in Tab. II. In each case, the pulse width is equal to the result from Tab. II, or smaller by ≈ 0.1 ns. The fits yield correlation coefficients that are smaller by 1.25% to 12% relative to the model including a reflected pulse. Thus, including the reflected pulse in the model improves the fit, but not to an extent that changes the interpretation of the event as a UHECR. For example, the authors of [15] showed that UHE-ν events from an ARA-like detector at the South Pole would produce a distribution of correlation coefficients peaked at $\rho \approx 0.9$ but spread between 0.4 and 1.0. If adding or removing the reflected pulse from the fit reduced ρ below 0.4, for example, this would suggest a shift in interpretation.

ID	$\sigma_{t,1}$ **(ns)**	$\sigma_{t,2}$ **(ns)**	Δt **(ns)**	**Rel. amp.**	ρ
1915-26288	0.6 ± 0.1	2.0 ± 0.1	25.25 ± 0.25	0.08 ± 0.01	0.86
1957-13330	0.5 ± 0.1	1.8 ± 0.1	11.5 ± 0.25	0.09 ± 0.01	0.74
2171-31805	0.6 ± 0.1	1.9 ± 0.1	25.25 ± 0.25	0.09 ± 0.01	0.81
2250-20189	0.7 ± 0.1	2.2 ± 0.1	24.75 ± 0.25	0.09 ± 0.01	0.81
2352-85489	0.7 ± 0.1	2.1 ± 0.1	24.75 ± 0.25	0.10 ± 0.01	0.84
2375-17342	0.6 ± 0.1	1.9 ± 0.1	19.0 ± 0.25	0.09 ± 0.01	0.73
2529-09767	0.7 ± 0.1	1.9 ± 0.1	25.5 ± 0.25	0.11 ± 0.01	0.82
2716-58611	0.8 ± 0.1	–	–	–	0.84
2782-00106	0.7 ± 0.1	2.1 ± 0.1	24.5 ± 0.25	0.11 ± 0.01	0.84
2955-47449	0.5 ± 0.1	1.6 ± 0.1	25.0 ± 0.25	0.03 ± 0.01	0.69
2961-98361	0.6 ± 0.1	1.9 ± 0.1	24.25 ± 0.25	0.05 ± 0.01	0.80
2978-29412	0.5 ± 0.1	1.7 ± 0.1	25.5 ± 0.25	0.05 ± 0.01	0.82
3352-89556	0.5 ± 0.1	1.8 ± 0.1	25.5 ± 0.25	0.05 ± 0.01	0.81

TABLE II: Results for the primary pulse width $\sigma_{\mathrm{t},1}$, reflected pulse width $\sigma_{\mathrm{t},2}$, delay between primary and reflected pulses Δt, reflected pulse amplitude (relative to primary), and correlation coefficient ρ.

ID	$\sigma_{t,1}$ **(ns)**	ρ	$\Delta\rho$
1915-26288	0.5 ± 0.1	0.83	-0.03
1957-13330	0.5 ± 0.1	0.66	-0.08
2171-31805	0.5 ± 0.1	0.775	-0.035
2250-20189	0.6 ± 0.1	0.80	-0.01
2352-85489	0.6 ± 0.1	0.83	-0.01
2375-17342	0.5 ± 0.1	0.69	-0.04
2529-09767	0.6 ± 0.1	0.80	-0.02
2716-58611	0.8 ± 0.1	0.84	–
2782-00106	0.6 ± 0.1	0.815	-0.025
2955-47449	0.6 ± 0.1	0.68	-0.01
2961-98361	0.6 ± 0.1	0.79	-0.01
2978-29412	0.5 ± 0.1	0.78	-0.04
3352-89556	0.5 ± 0.1	0.78	-0.03

TABLE III: Same as Tab. II, but with no reflected pulse in the model. The term $\Delta\rho$ refers to the decrease in correlation coefficient listed in Tab. II.

V. CONCLUSION

An example of what the analysis reveals about the UHECR candidate data is shown in Fig. 7. In Fig. 7 (left) the CSW fit from Fig. 4 (bottom right) is shown.

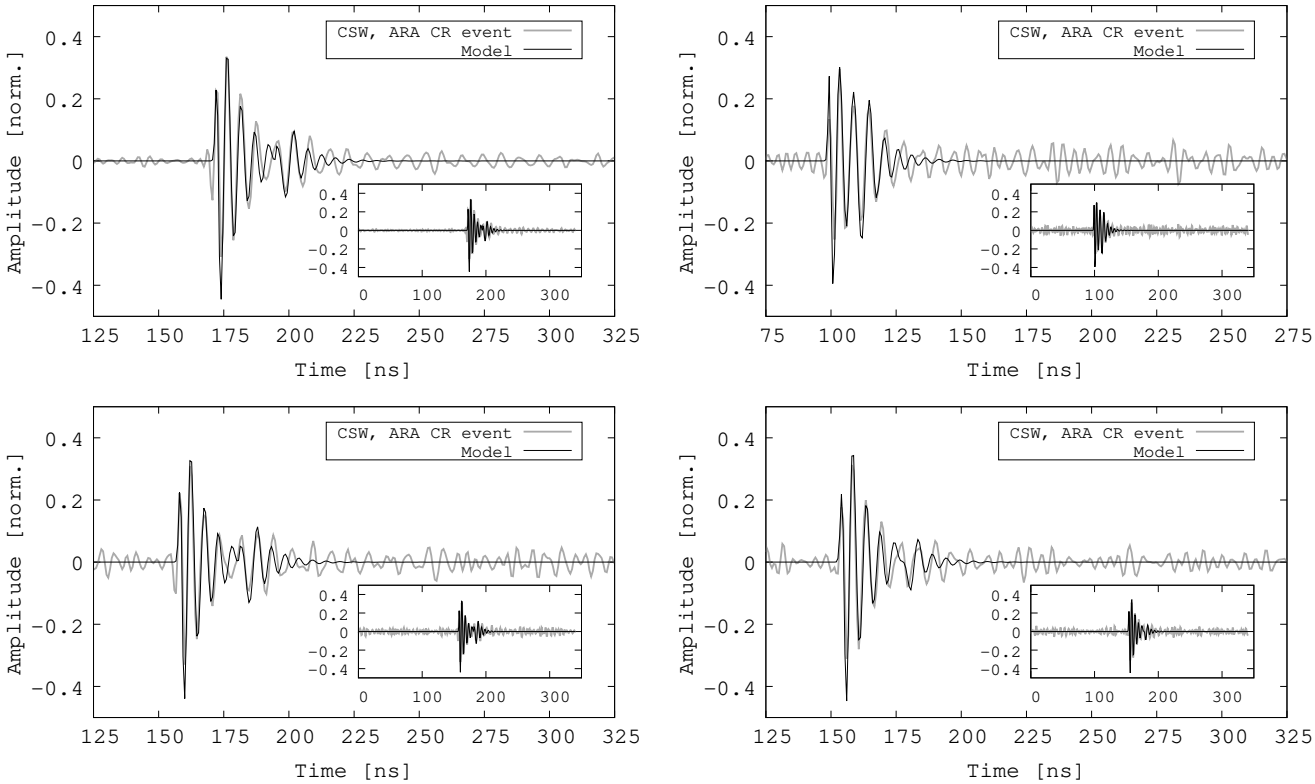

FIG. 3: UHECRs, by ID: (top left) 1915-26288, (top right) 1957-13330, (bottom left) 2171-31805, (bottom right) 2250-20189.

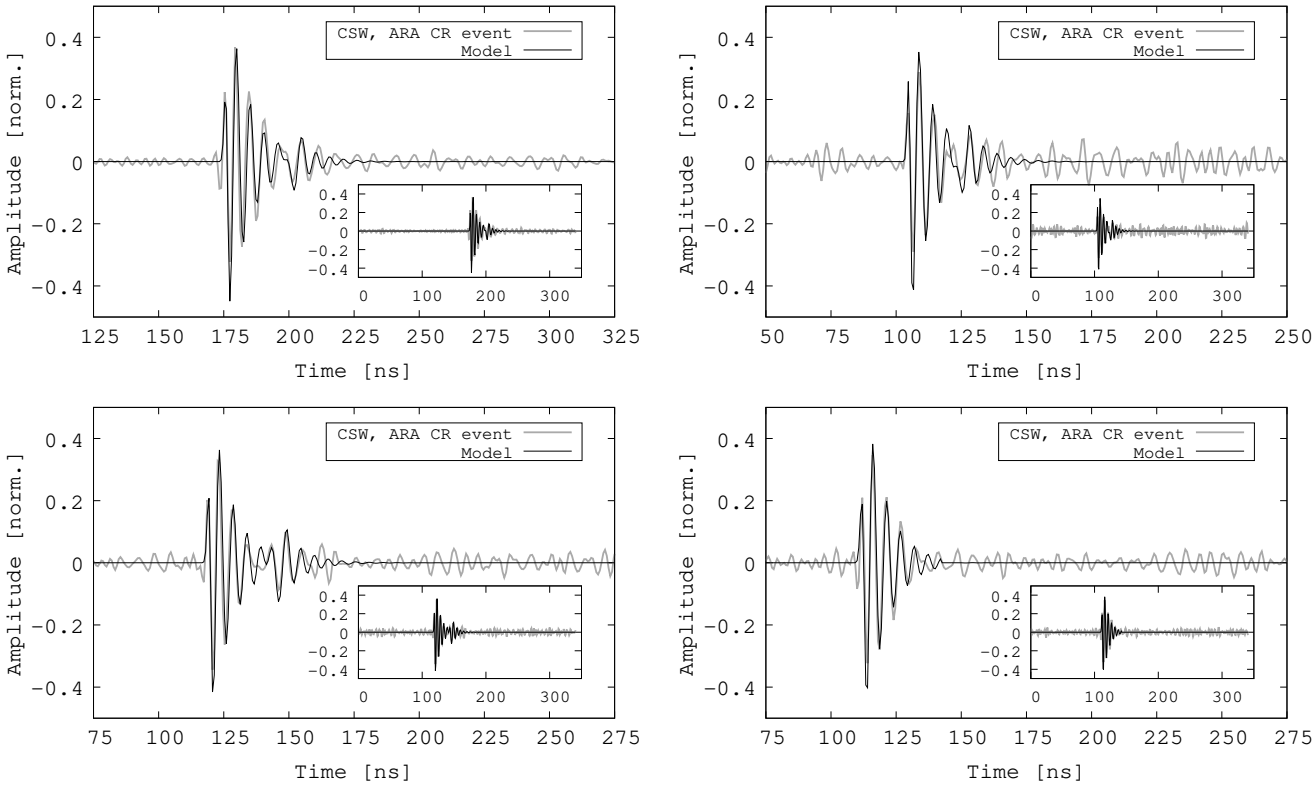

FIG. 4: UHECRs, by ID: (top left) 2352-85489, (top right) 2375-17342, (bottom left) 2529-09767, (bottom right) 2716-58611.

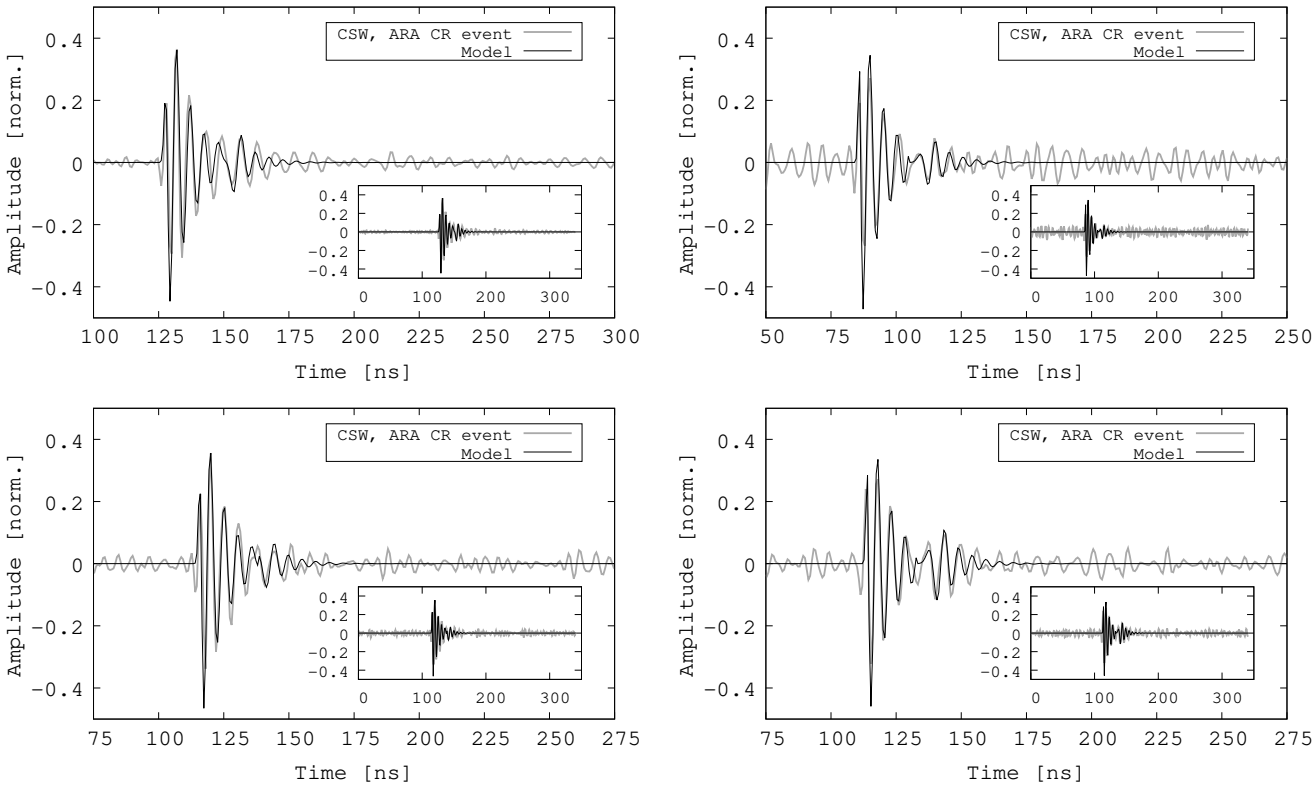

FIG. 5: UHECRs, by ID: (top left) 2782-00106, (top right) 2955-47449, (bottom left) 2961-98361, (bottom right) 2978-29412.

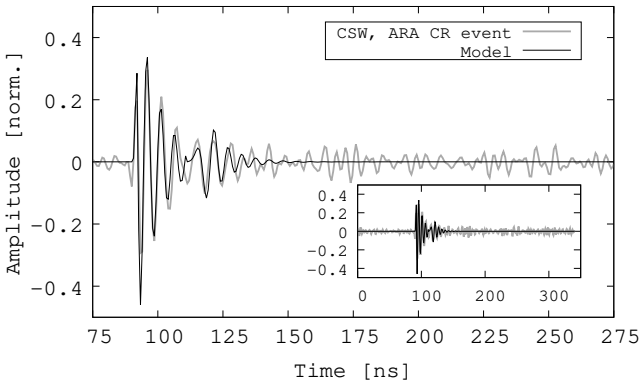

FIG. 6: UHECRs, by ID: 3352-89556

This CSW did not contain a reflected pulse, and the correlation between model and data was 0.84. The reconstructed σ_t was 0.6 ± 0.1 ns. Knowing σ_t allows us to graph the reconstructed $\vec{E}$-field, normalized with $E_0 = 1$ in Eq. 1. The field is graphed in Fig. 7 (right), for $\sigma_t = 0.6$ ns, $\sigma_t = 0.5$ ns, and $\sigma_t = 0.7$ ns, reflecting the uncertainty in σ_t. The reconstructed $\vec{E}$-field from our analysis matches the central portion of the blue data shown in Fig. 2 of [4]. The data was taken from Fig. 2 of [4], smoothed with a 10-sample running average filter, and inverted. Our analytic reconstruction technique (fitting Eq. 7 to CSWs from raw, observed voltage traces) produces an $\vec{E}$-field field that matches the deconvolution used to produce the reconstructed $\vec{E}$-field in Fig. 2 of [4].

Deconvolution involves several general stages. The first step is to measure the impulse response $r(t)$ of the entire RF detection chain in an anechoic chamber or RF bench testing environment. Once the data from the deployed detector is collected, the second stage involves filtering it for noise. Thermal noise has a detrimental effect on deconvolution algorithms [22]. The final step is to use techniques like forward-folding or the convolution theorem to calculate the $s(t)$ that causes $s(t) * r(t)$ to match the data. The three-stage general process outlined here con-

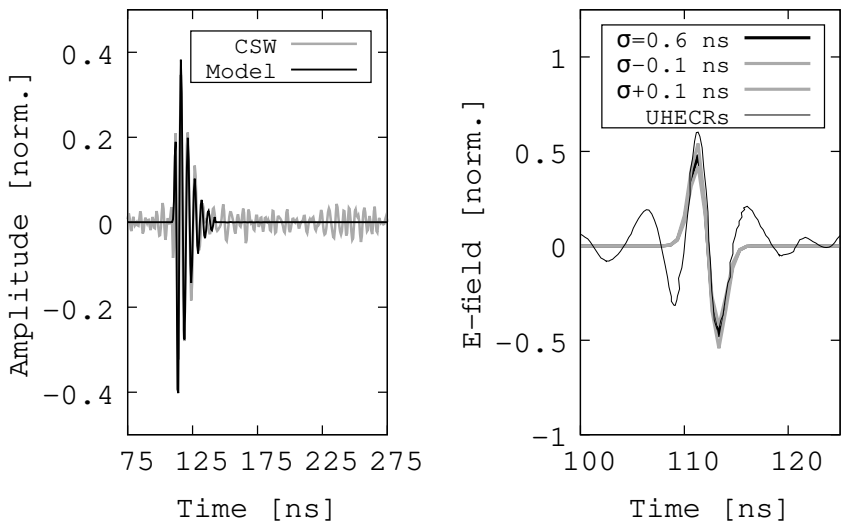

FIG. 7: (Left) the results from Fig. 4 (bottom right). (Right) Equation 1, evaluated with the reconstructed σ_t from Tabs. II and III, and $E_0 = 1$. Thick black line: $\sigma_t = 0.6$ ns. Thick gray lines: $\sigma_t = 0.7$ ns and $\sigma_t = 0.5$ ns. Thin black line: average data from UHECR $\vec{E}$-fields, adapted from Fig. 2 of [4].

tains opportunities for error propagation, and the match in Fig. 7 (right) demonstrates the algorithms used to calculate the $\vec{E}$-field agree with first principles.

Finally, measurements of σ_t from UHECR candidates can be used to calculate parameters of the UHECR cascade observation. Consider Eq. 5 of [15]:

$$a\Delta\theta = \frac{c\sigma_t}{\sin\theta_{\mathrm{C}}} \tag{9}$$

In Eq. 9, a is a measure of the longitudinal length of the in-ice cascade in meters, $\Delta\theta$ is $\theta - \theta_{\mathrm{C}}$, where θ is the viewing angle with respect to the cascade axis, and θ_{C} is the Cherenkov angle, c is the speed of light in ice, and σ_t is the pulse width. The authors of [14] calculated for 10-100 PeV cascades, $a \approx 5$ m. Using $\sigma_t = 0.6$ ns, and $c = 0.3/1.78$ m ns^{-1} and $\theta_{\mathrm{C}} = 55.8$ degrees for RF waves in ice, we find $\Delta\theta \approx 1.4$ degrees. This result aligns with observed UHECR candidate power spectra shown in Fig. 2 of [4]. The red band in Fig. 2 of [4] is calculated assuming $\Delta\theta \leq 3$ degrees. Assuming that the UHECR cascade energy was between 10 and 100 PeV gives the correct range for $\Delta\theta$.

There are two logical steps for this work in the future. First, a separate fit for $\Delta\theta$ would allow the measrurement of the a parameter, which is tied to the UHECR cascade energy. This parameter $\Delta\theta$ may be constrained by using the spectral cutoff of the voltage traces, which are inversely correlated to $\Delta\theta$ [13]. Second, it is interesting to note that our model for $s(t)$ in Eq. 1 corresponds to the *off-cone* ($\theta \neq \theta_{\mathrm{C}}$) result from HH2022 [14]. The authors of [14] also include an *on-cone* version of $s(t)$ that is used when $\Delta\theta = 0$, and can be used as $\Delta\theta \to 0$. The on-cone model could provide a better fit for events with $\Delta\theta \lesssim 1$ degree [14]. Finally, it is our hope that one day our model will be used to fit UHE-ν candidates, in addition to UHECR candidates.

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